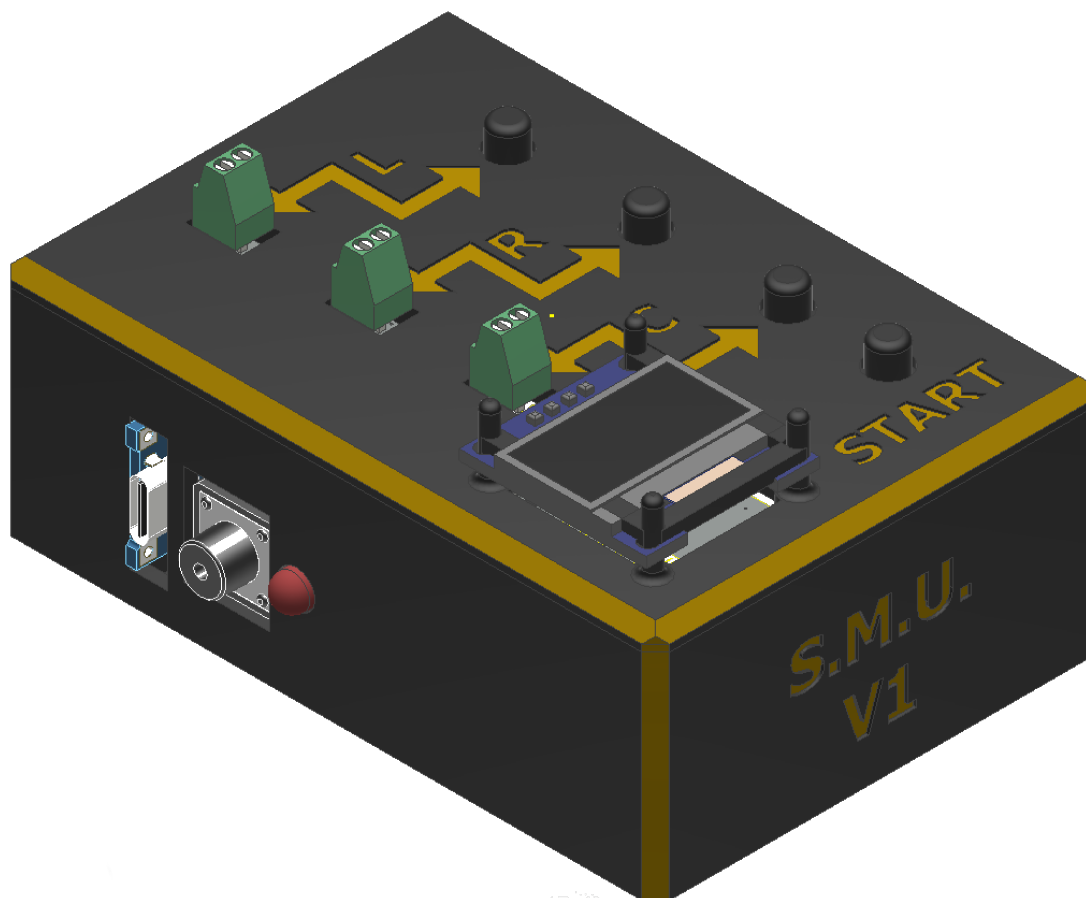
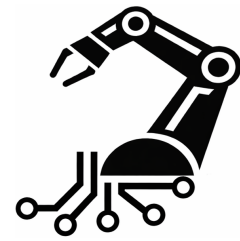


S.M.U. Project

Final prototype documentation



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Revision	Rev. A
Date	January 25, 2026
Version	1.0

This document summarizes the prototype stage of the S.M.U. device and describes its design assumptions, construction and operation.

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Concept

1.1 Device idea and purpose

The aim of the project was to create a meter capable of measuring typical electronic components used in mechatronics. The version described in this document is a first prototype, built to check whether the selected measurement methods work in practice and to assess the possibility of developing a complete device. This part of the project focused mainly on the measurement section, while a ready-made microcontroller board was used for calculations and result interpretation. The Raspberry Pi Pico was selected for this project because it was inexpensive and allowed testing a different family of microcontrollers.

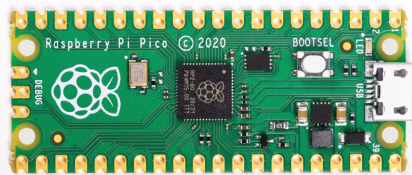


Figure 1.1: Selected microcontroller module.

1.2 Resistance measurement

Resistance measurement is based on a voltage divider and a reference resistor. The value of the measured resistor is calculated from the voltage measured by the microcontroller ADC.

The reference resistor values were selected to keep the ADC reading in a useful part of its input range for different measured resistances.

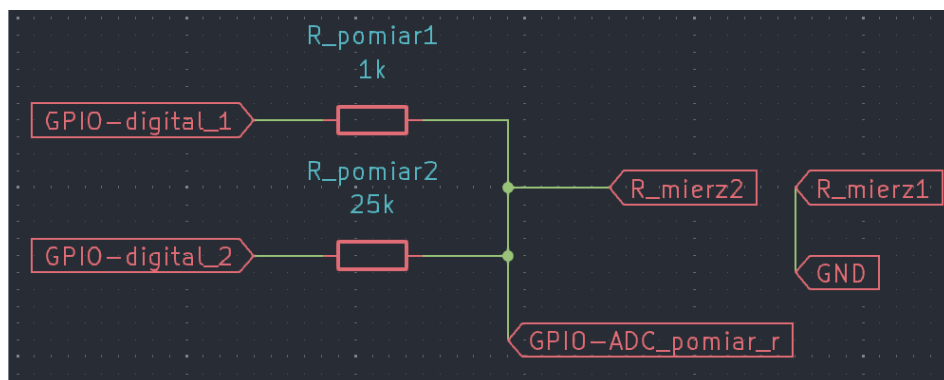


Figure 1.2: Resistance measurement circuit diagram.

1.3 Capacitance measurement

Capacitance measurement was based on the time response of an RC circuit. The component values and the measurement method were first checked in simulation. The capacitance is calculated by measuring the capacitor discharge time and using the RC time constant τ .

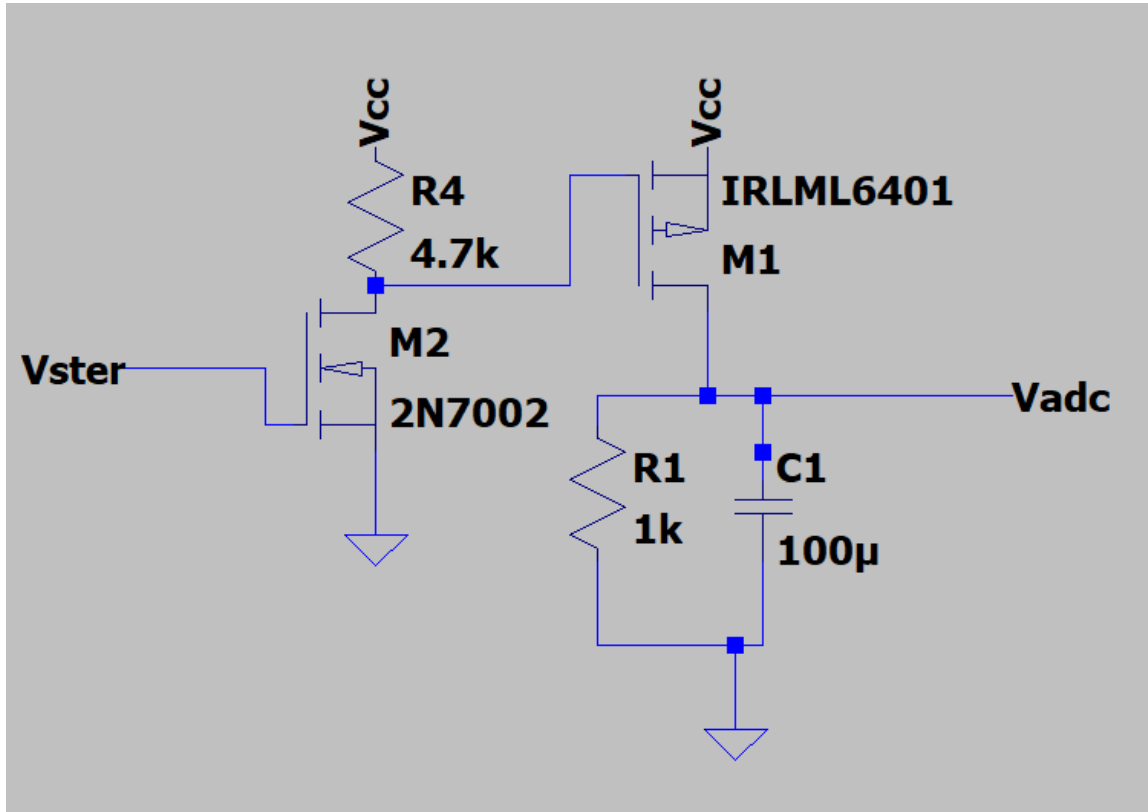


Figure 1.3: Simulation circuit for capacitance measurement.

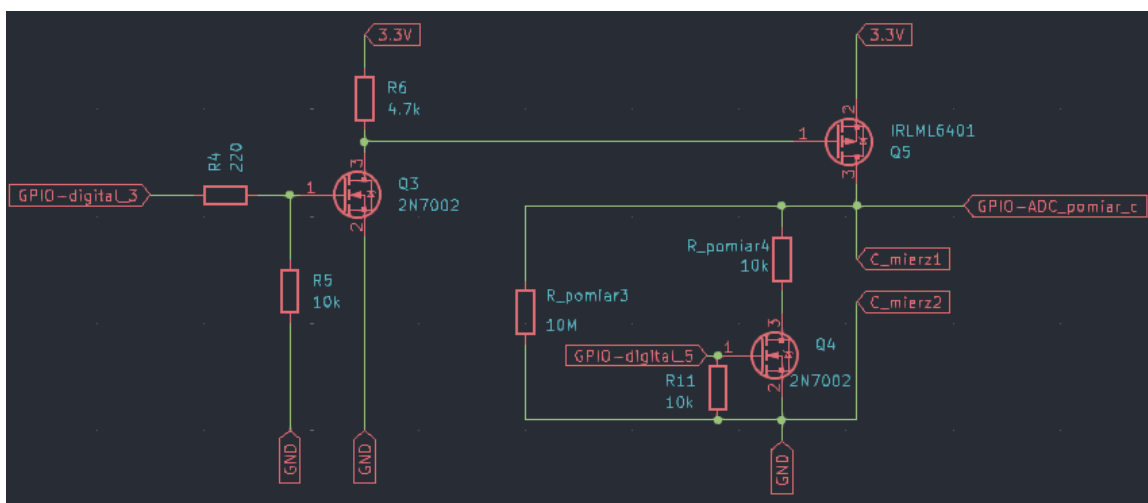


Figure 1.4: Capacitance measurement circuit diagram.

The capacitance calculation is based on measuring the time after which the capacitor voltage drops to about 37% of its initial value. This point corresponds approximately

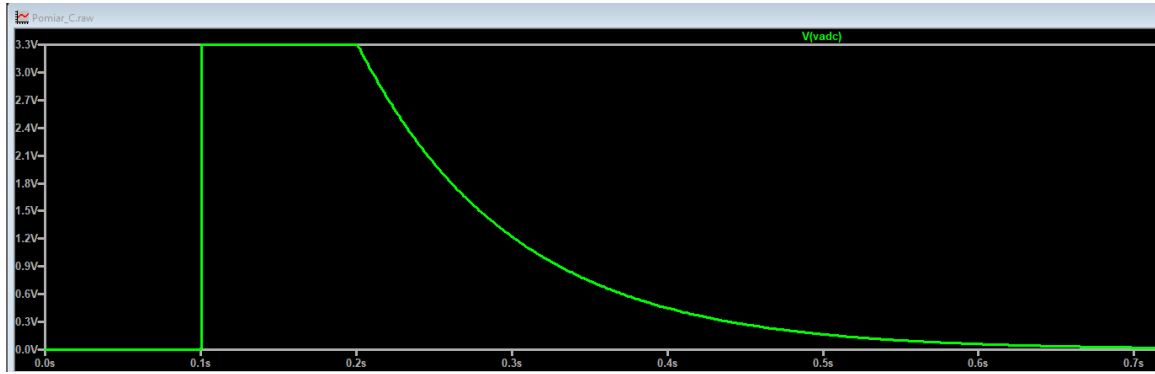


Figure 1.5: Simulation result of the measurement circuit.

to one time constant. Two measurement resistors were used to cover a wider range of capacitances. For large capacitances and a high resistance value, the measurement time could become too long. For small capacitances and a low resistance value, the discharge could be too fast for reliable measurement by the microcontroller. For this reason, an additional parallel resistor was added and switched by an N-MOSFET. This reduced the measurement resistance from 1 M Ω to about 10 k Ω when needed.

1.4 Inductance measurement

Inductance measurement was based on the resonant frequency of an LC circuit. Before building the practical circuit, simulations were performed to check the adopted measurement idea. The measured inductance is calculated from the resonant frequency of the LC circuit, using a known capacitance value.

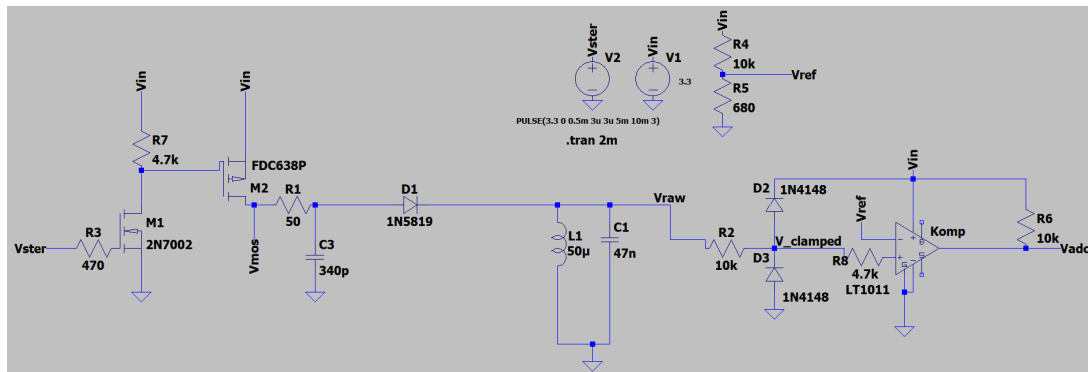


Figure 1.6: Simulation circuit for inductance measurement.

In the first version of the prototype, an amplifier was used between the resonant circuit and the comparator. After testing, it was removed because the circuit could work without it. The amplifier was originally intended to increase the amplitude of the signal from the resonant circuit. Schottky diodes were used as clamps to limit the negative part of the signal and allow the comparator to work correctly. The comparator converts the analog oscillating signal into a digital signal that is easier for the microcontroller to count.

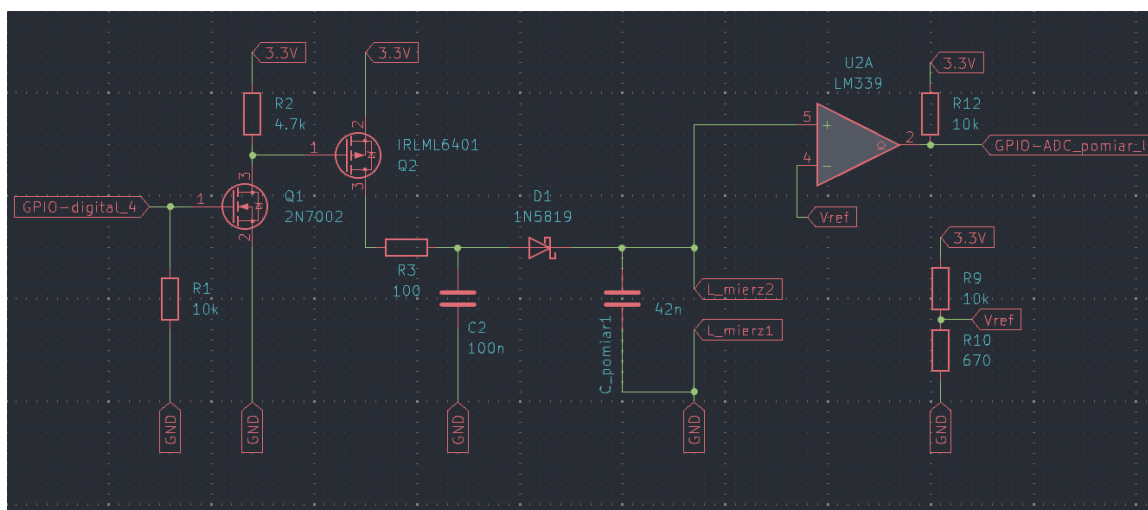


Figure 1.7: Inductance measurement circuit diagram.

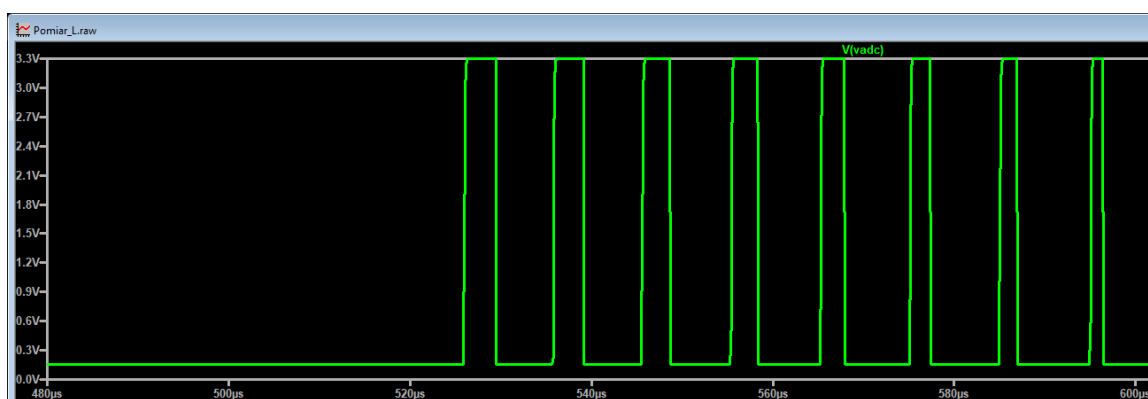


Figure 1.8: Simulation result of the measurement circuit for $L = 50 \mu H$.

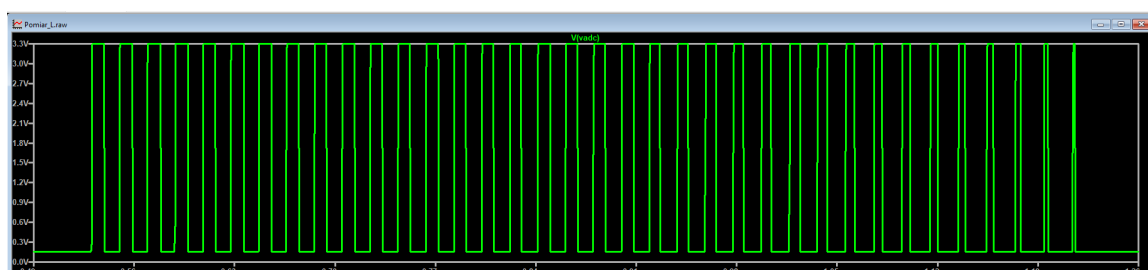


Figure 1.9: Simulation result of the measurement circuit for $L = 200 \mu H$.

1.5 Power supply section

The project included USB power supply and the possibility of charging a battery cell. The power supply section was designed to make the device easy to use and allow further development.

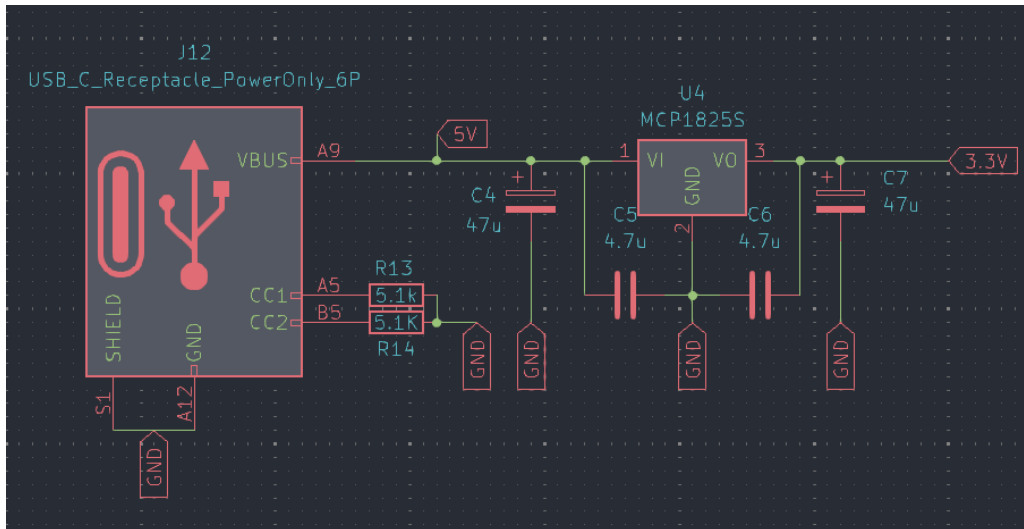


Figure 1.10: USB power supply handling circuit diagram.

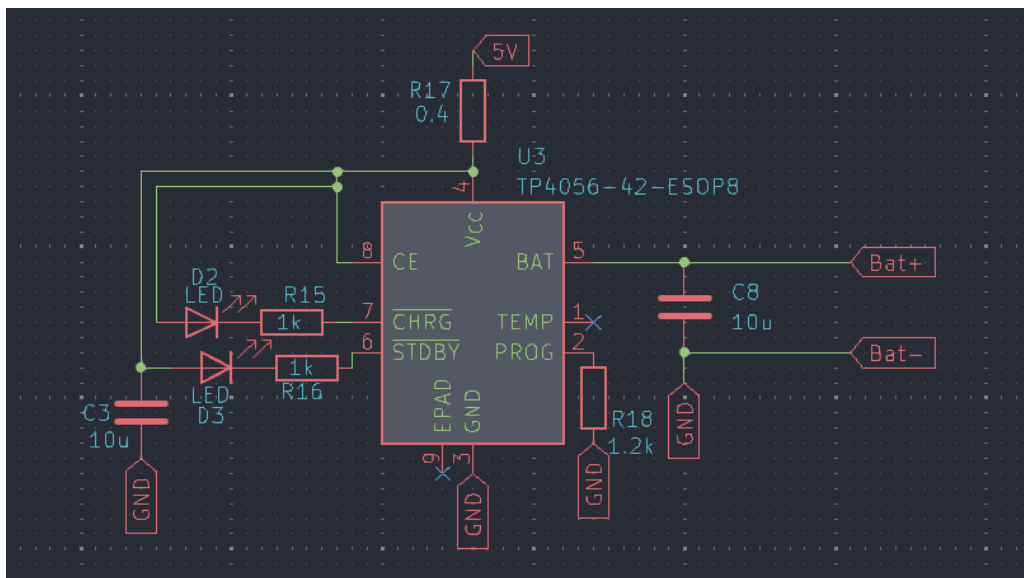


Figure 1.11: Battery cell charging circuit diagram.

Later, it was noticed that the TP4056 chip requires the “TEMP” pin to be connected to GND when this function is not used. More importantly, the chip is responsible only for the charging process and does not provide full battery protection. In this configuration, the circuit was not fully safe, because there was no protection system monitoring the battery cell. Such protection circuits usually use a pair of back-to-back N-MOSFETs to disconnect the cell from the system when an unsafe condition is detected. After analyzing this problem, the USB and TP4056 section was not assembled. Instead, a ready-made charging module with battery protection was used.

1.6 PCB

Based on the prepared schematic, a printed circuit board was designed. The top and bottom sides of the PCB are shown below.

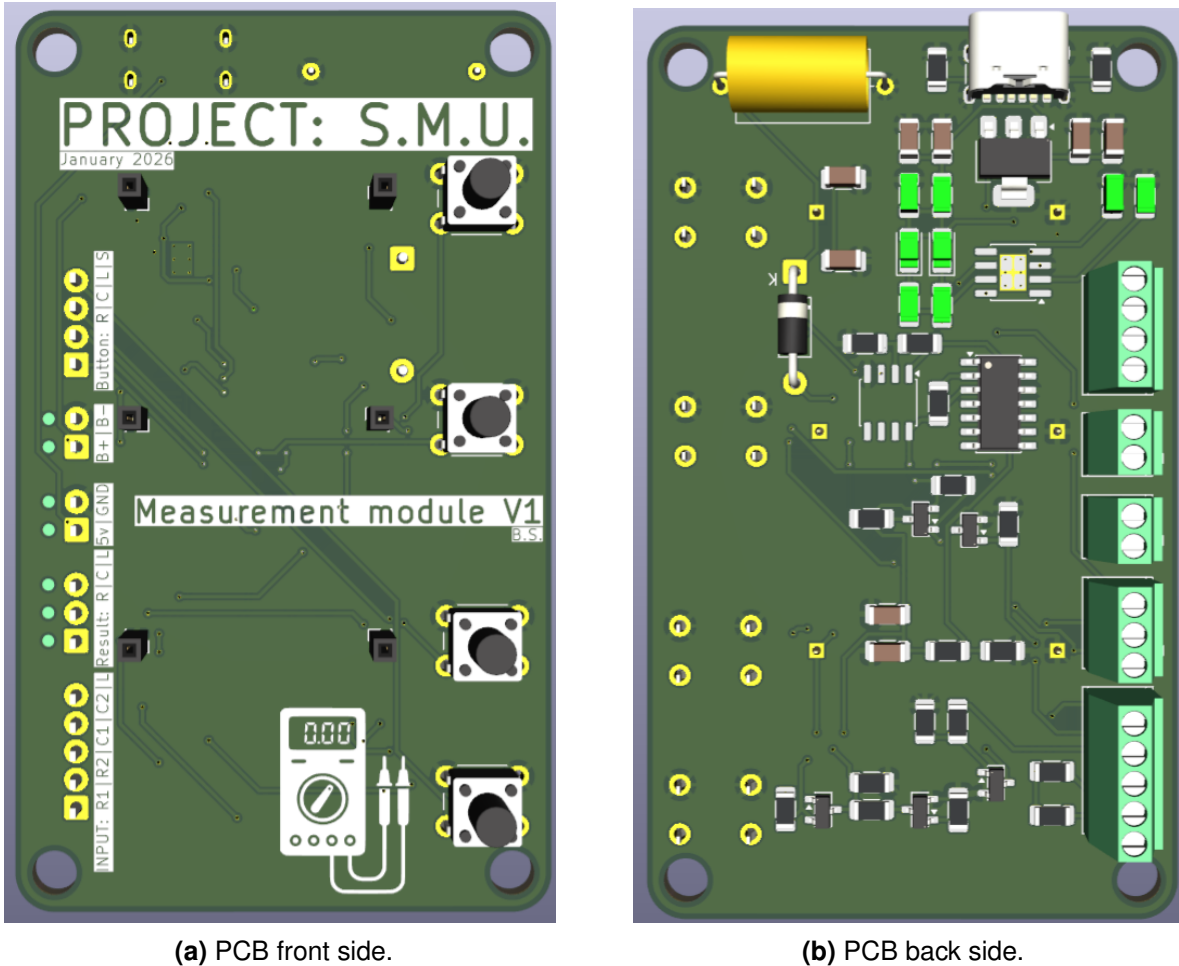


Figure 1.12: View of the designed PCB.

Compared with the original PCB design, the final assembled version does not include the TP4056 circuit or the amplifier, for the reasons described earlier. In addition, clamp diodes were added to ensure proper operation of the comparator. Another change was related to the measuring capacitor in the resonant path used for inductance measurement. A precision film capacitor with a different pin layout than the one used in the original design was installed.

Application

2.1 Accuracy

To evaluate the operation of the device, a series of five measurements was performed for selected components. For resistance and capacitance, the results were compared with the average value obtained from a reference meter. For inductance, due to the lack of reference meter data, the results were compared with the nominal values of the tested coils.

2.1.1 Resistance

Results table

Table 2.1: Comparison of resistance measurement results

Element	Average value from reference meter	Average value from S.M.U.	Error relative to reference
100 Ω	97.64 Ω	96.50 Ω	-1.17%
1 k Ω	972.24 Ω	956.23 Ω	-1.65%
47 k Ω	46.962 k Ω	44.032 k Ω	-6.24%
100 k Ω	96.372 k Ω	87.646 k Ω	-9.05%
1 M Ω	1.0036 M Ω	0.5204 M Ω	-48.15%

Description

Based on the measurements, the circuit correctly measures resistance in the lower and middle part of the tested range. For the 100 Ω and 1 k Ω resistors, the deviation from the reference value did not exceed 2%. As the measured resistance increased, the error also increased, which was visible for 47 k Ω and 100 k Ω . The largest deviation occurred for the 1 M Ω resistor, where the measured value was almost two times lower than the reference value.

Conclusions

The resistance measurement circuit can be considered suitable for low and medium resistance values. In the current prototype version, the biggest problems appear at high resistances, where ADC resolution, reference resistor tolerance and noise have a much stronger influence on the result. This section requires further improvement if the device is to measure values in the range of hundreds of kilohms and megaohms reliably.

Theoretical and practical range

The theoretical resistance measurement range results directly from the voltage divider relation:

$$U_{ADC} = V_{CC} \frac{R_x}{R_x + R_w}$$

where R_x is the measured resistance and R_w is the reference resistor. In practice, the most useful ADC operating area is not at the very beginning or end of the scale, but rather in the middle part of the characteristic. With a simple assumption that, for more accurate calculation, it is convenient to keep the ADC input voltage in the range of about 10%–90% of full scale, the following relation is obtained:

$$0.1 \leq \frac{R_x}{R_x + R_w} \leq 0.9$$

which after transformation gives approximately:

$$0.11 R_w \leq R_x \leq 9 R_w$$

For an example reference resistor $R_w = 10 \text{ k}\Omega$, this gives a range of about:

$$1.1 \text{ k}\Omega \leq R_x \leq 90 \text{ k}\Omega$$

which is approximately from about $1 \text{ k}\Omega$ to $100 \text{ k}\Omega$ for more accurate calculation. With a wider but less favorable use of the ADC range, for example about 5%–95%, the approximate range would be from about $0.5 \text{ k}\Omega$ to about $190 \text{ k}\Omega$. The actual results showed that in the current prototype version the useful range mainly covered values from about 100Ω to about $100 \text{ k}\Omega$, while for $1 \text{ M}\Omega$ the error was already too large to consider this point reliable.

2.1.2 Capacitance

Results table

Table 2.2: Comparison of capacitance measurement results

Element	Average value from reference meter	Average value from S.M.U.	Error relative to reference
1000 μF	1072.6 μF	1036.0 μF	-3.41%
470 μF	481.38 μF	439.93 μF	-8.61%
100 μF	124.24 μF	102.14 μF	-17.79%
10 μF	11.56 μF	9.316 μF	-19.41%
100 nF	146.03 nF	32.08 nF	-78.03%

Description

For capacitance measurement, correct and repeatable results were obtained for larger capacitances, especially in the range of 470 μF –1000 μF . As the capacitance decreased, the measurement error increased. For 100 μF and 10 μF capacitors, the result was clearly lower than the reference value. For 100 nF, the measurement was incorrect. This shows that the method works well for larger time constants, but for small capacitances parasitic effects and timing limitations become dominant.

Conclusions

The capacitance measurement section works correctly for capacitors with larger values, while the lower range requires further optimization. The results suggest that in the current configuration the circuit is more useful for measuring electrolytic capacitors and larger capacitors than for precise measurement of small capacitances. In particular, for the nanofarad range, changing the measurement components or using another method should be considered.

Theoretical and practical range

For capacitance measurement, the starting point was the time constant relation:

$$\tau = RC$$

where the measurement was based on determining the time required for the voltage to drop to about 37% of the initial value, which corresponds to one time constant τ . Two measuring resistors were used in the project: about 1 $M\Omega$ and about 10 $k\Omega$, with the second value obtained by connecting a parallel branch. Therefore, the theoretical range can be estimated by assuming that a convenient measurement time for the microcontroller is roughly between 10 ms and 10 s .

For $R = 1 M\Omega$:

$$C = \frac{\tau}{R}$$
$$C_{min} \approx \frac{10 ms}{1 M\Omega} = 10 nF$$
$$C_{max} \approx \frac{10 s}{1 M\Omega} = 10 \mu F$$

For $R = 10 k\Omega$:

$$C_{min} \approx \frac{10 ms}{10 k\Omega} = 1 \mu F$$
$$C_{max} \approx \frac{10 s}{10 k\Omega} = 1000 \mu F$$

This gives a total theoretical range of about 10 nF to about 1000 μF , where the lower part of the range is mainly covered by the 1 $M\Omega$ resistor and the upper part by the resistor of about 10 $k\Omega$. In practice, the obtained results showed that the current prototype version worked correctly mainly from about 10 μF to about 1000 μF . Although simple

calculations suggest that lower values should also be possible, for small capacitances parasitic capacitances, component tolerances and timing limitations of the circuit became increasingly important. As a result, for 100 nF the error was already very large.

2.1.3 Inductance

Results table

Table 2.3: Comparison of inductance measurement results

Element	Nominal value	Average value from S.M.U.	Error relative to nominal value
330 μH	330 μH	329.87 μH	-0.04%
220 μH	220 μH	170.09 μH	-22.69%
100 μH	100 μH	106.27 μH	+6.27%

Description

For inductance measurement, the results were mixed. For the 330 μH coil, the measured value was very close to the nominal value, which supports the measurement idea based on the resonant method. For 100 μH , the deviation was still acceptable. The biggest problem appeared for 220 μH , where the result was significantly lower than the nominal value. This may be caused by parasitic parameters of the tested coil, limited frequency counting stability or a faulty coil sample.

Conclusions

Inductance measurement can be considered promising, but in the current version it is not yet fully stable across the entire tested range. The very good result obtained for 330 μH shows that the circuit has potential, but the error spread for the other coils suggests the need for further calibration and better consideration of parasitic elements. In practice, this means that this section still requires improvement before it can be considered fully universal.

Theoretical and practical range

For inductance measurement, the relation for the resonant frequency of an LC circuit was used:

$$f = \frac{1}{2\pi\sqrt{LC}}$$

after transformation:

$$L = \frac{1}{(2\pi f)^2 C}$$

Assuming a reference capacitor $C_{ref} = 42\text{ nF}$, it is possible to estimate what inductance range corresponds to frequencies convenient for further counting and processing by the

circuit. For an approximate frequency range from about 40 kHz to 110 kHz, the following values are obtained:

for $f = 110 \text{ kHz}$:

$$L \approx \frac{1}{(2\pi \cdot 110000)^2 \cdot 42 \cdot 10^{-9}} \approx 50 \mu H$$

for $f = 40 \text{ kHz}$:

$$L \approx \frac{1}{(2\pi \cdot 40000)^2 \cdot 42 \cdot 10^{-9}} \approx 377 \mu H$$

This gives a theoretical operating range of about:

$$50 \mu H \leq L \leq 380 \mu H$$

It is worth noting that for values close to the tested ones, frequencies consistent with the idea of this method are obtained: for 100 μH about 78 kHz, for 200 μH about 55 kHz, and for 330 μH about 43 kHz. This means that the adopted range from several dozen to several hundred μH was theoretically justified. In practice, values from 100 μH to 330 μH were tested, and the best agreement was obtained for 330 μH , while for 220 μH a clear error appeared. This shows that the method itself was correct, but the practical range of accurate measurement turned out to be narrower than the range resulting from a simple theoretical model.

Summary of the measurement section

The measurements confirmed that the adopted measurement methods can work in practice, but they also showed that the device accuracy strongly depends on the tested value range. The best results were obtained in the middle parts of the measurement ranges, where the values were close to the reference values and showed good repeatability. The largest deviations appeared at extreme values. This indicates the influence of method limitations, parasitic component parameters and the resolution of the measurement path. The prototype fulfilled its verification role and helped identify the areas that require further improvement.

Direction of further development

After analyzing the results, it can be concluded that the measurement circuit concept is valid. However, the results also show that in the next version it is necessary to:

- carefully analyze the measurement ranges and add more measurement elements,
- remove the ready-made charging module,
- add protection circuits for the battery cell,
- replace the ready-made microcontroller module with a dedicated microcontroller circuit,
- analyze available microcontrollers and choose one with a suitable ADC, or use an external ADC,
- consider a better physical connector for the measured components,
- add protection against reverse connection of electrolytic capacitors.